E-Commerce Customer Journey Analytics

Analytical Project Report | MySQL · Power BI · DAX

Data period: 24 September 2020 – 28 February 2021. Dashboard values below are rounded display figures.

Executive Summary

This project examines product browsing, cart activity, purchases, and revenue in a historical electronics e-commerce event dataset. MySQL was used to prepare, validate, and analyze event records; Power BI was used to develop an interactive dashboard for exploring purchasing activity and product performance.

Business Objectives

Assess the distribution of view, cart, and purchase events; compare products, brands, and categories; examine revenue patterns over time; and enable filtering by date, category, brand, and event type.

Dashboard KPIs

Metric	Displayed value
Total product views	~793K
Total cart events	~54K
Total purchase events	~37K
Total purchase revenue	~5.13M
Average purchase-event value	137.24

Revenue currency is not independently verified. Purchase-event value should not be assumed to equal full order value.

Methodology

Data preparation. Imported records into MySQL and inspected missing values, repeated records, timestamps, and nonpositive prices. Standardized text and date fields with TRIM, NULLIF, REPLACE, and STR_TO_DATE; maintained raw, cleaned, and analysis-ready tables.

SQL analysis. Calculated event counts, distinct users and sessions, purchase-based revenue, average purchase-event value, top products, and daily revenue using GROUP BY, conditional aggregation, COUNT(DISTINCT), and revenue reconciliation.

Power BI. Created DAX measures and interactive views for KPIs, event-stage comparison, event distribution, monthly revenue, category and brand revenue, and top products. Tested slicer interactions and confirmed displayed KPI totals.

Key Findings and Interpretation

Activity mix. Product views represent most recorded events. Approximately 54K cart events and 37K purchase events appear alongside ~793K views. The purchase-to-view event ratio is approximately 4.7%, but it is **not** a verified customer or session conversion rate.

Revenue. Recorded purchase-event revenue totals approximately 5.13M, with an average value of 137.24 per purchase event. Brand, category, and product comparisons support further investigation of revenue concentration.

Time. Revenue varies across recorded months. September and February contain partial coverage, so their raw totals should not be interpreted as directly comparable full-month growth.

Business Recommendations

Investigate the customer journey with session-ordered events before drawing conclusions about conversion drop-off. Examine products with high view activity but relatively few purchase events; monitor revenue-driving products and brands; and improve completeness of category and brand attributes to support reliable segmentation.

Limitations

Event counts may include repeated actions by the same user. The funnel compares event types rather than confirmed sequential stages. Exact-row deduplication can remove legitimate repeated events when unique event IDs are unavailable. Purchase measures may appear blank when filtering to view-only events. KPI sparklines are decorative rather than calculated historical trends. Product names and currency are not inferred when absent from the data.

Conclusion

The analysis provides a structured view of e-commerce engagement and purchase-related revenue through a reproducible SQL workflow and an interactive Power BI dashboard. It highlights the predominance of browsing activity, differences in product and brand performance, and opportunities to improve data completeness and investigate the purchasing journey. The findings provide a foundation for further analysis; stronger conversion conclusions would require session-level event sequencing and additional behavioral context.